Research Notes Spring 2014

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1 Non Consecutive Congruences

In the paper by Blair, Blass, and Howard [BBH] it is proved that there can be Dedekind-finite sets which are congruent to only the elements of some specified set of consecutive congruence classes. The paper by John Holler and Matt Tanzer [HT] proved that there can be Dedekind-finite sets which are members of two non-consecutive congruence classes. The natural next question is can a Dedekind-finite set be congruent to all of a set of non-consecutive congruence classes when that set has more than two members. I will begin by showing that there can be a Dedekind-finite set congruent to 0, 2, and 4 (mod 6) but not to 1, 3, or 5.

Theorem 1.1. *It is consistent that there exists a Dedekind-finite set X, such that* $X \equiv 0, 2, 4$*(mod 6) but* $X \not\equiv 1, 3, 5$*(mod 6).*

Proof. Let A be a countably infinite set of atoms. Let P be a partition of A into pairs and $n \mapsto P_n$ be an enumeration of those pairs. Let M be the permutation model resulting from allowing only the automorphisms which fix every P_n. This set can be partitioned into pieces of size 6 by breaking it into consecutive triplets of pairs. It is easy to see that if one or two pairs are removed, the set can still be partitioned into pieces of size 6.

Assume, toward a contradiction, that $X - a$ is divisible by 6 for $a = 1, 3, 5$ by some partition Q. Assume $E = \bigcup_{i=0}^{N} P_n$ supports Q. Then for any pair outside of E, there is a permutation which transposes that pair while fixing everything else. Thus any Q block which is not touching E is made up of 3 pairs. Therefore the finite set consisting all Q blocks touching E and the

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elements to be removed is made up of pairs. Thus this finite set has even size, yet, when we remove an odd number of elements, the set still has an even size because it is divisible by 6.

□

While this proof does tell us that sets can be equivalent to more than two non-consecutive classes, it tells us nothing about cases with more than three congruence classes. Luckily the theorem is easily generalized to arbitrarily large collections of non-consecutive congruence classes. The generalization also allows for the gap between congruence classes to be larger than 1.

Theorem 1.2. *If a divides d and $2 \leq a < d$, then it is consistent to have Dedekind finite X such that $X - r$ is divisible by d if and only if r is a multiple of a.*

Proof. Let A be a countable set of atoms and let P be a partition of A into blocks of size a with an enumeration $n \mapsto P_n$. The permutations of A will be those which preserve every P_n. Let M be the resulting permutation model. If we group together d/a consecutive P_ns we will get a partition into pieces of size d. Clearly if we remove any number of P_ns we can still do this. So $X - ka$ is divisible by d for all $k \in \omega$. A is clearly Dedekind-finite.

Now assume we remove r elements from the first d/a P_ns to form $A - R$. Assume Q is a partition of $A - R$ into pieces of size d. Let $E = \bigcup_{i=0}^{N} P_n$ be a support of Q where $N \geq d/a$.

If z and z' are in the same P block (outside of E), there is a permutation which transposes z and z' but fixes everything else (including E). Therefore z and z' must be in the same Q block. So any P block outside of E is completely contained in exactly one Q block.

Now consider the union of the P blocks which are either part of E or in the same Q block as something in E. This finite set is divisible by a since it is a union of P blocks. But after we remove R it is divisible by d because of Q and thus divisible by a. So there is some finite n such that a divides n and a divides $(n - r)$. This is only possible if r is a multiple of a. □

This tells us that we can have sets which are congruent to many non-consecutive elements. It requires however that the gaps between the congruence classes be equal in size. The next step would be to see if there can exist X such that $X \equiv 0, 2, 3 (\text{mod } 5)$.

Theorem 1.3. *It is consistent to have Dedekind-finite Y such that Y, $Y - 2$, and $Y - 4$ are all divisible by 5 but $Y - 1$ and $Y - 3$ are not.*

Proof. (extended from proposition 3 of [HT]) Let Y be the disjoint union of two countably infinite sets of atoms. Call them X and X^*. Assign to them P, P^*, Q, and Q^* as described in proposition 3. Let M be the permutation model generated by the permutations which fix P, P^*, Q, and Q^*. Y is Dedekind finite because X and X^* are both Dedekind-finite. $P \cup P^*$ divides Y by 5. $P \cup Q^*$ divides $Y - 2$ by 5. $Q \cup Q^*$ divides $Y - 4$ by 5.

To show that $Y - a$ is not divisible by 5 for $a = 1, 3$, assume that $E = \bigcup_{i=0}^{N}(P_i \cup P_i^*)$ supports some partition $\mathcal{S}$ of $Y - a$ into pieces of size 5. Both of the claims on page 6 of [HT] are still true in this case. Also for the same reason as the second claim we have that a pair in X can not be in the same $\mathcal{S}$ block as a distinguished element from X^*. With these facts the rest of the proof follows in the same way as Proposition 3. □

Corollary 1.4. *There exists Dedekind-finite X such that $X \equiv 0, 2, 3$ (mod 5).*

Proof. $X = Y - 2$ where Y is the set from theorem 1.3. □

We are now very close to knowing that there can be a Dedekind finite set equivalent to any collection of natural numbers (mod 5). The only case that we have not proven is when $X \equiv 0, 3$ (mod 5) and nothing else. The proof for this is a modification of Proposition 3 from [HT].

Note: The following theorem can be proved in one line using sets described in [HT]. The presented proof, however, leads naturally into the proof of 1.6.

Theorem 1.5. *It is consistent to have Dedekind-finite X such that $X \equiv 0, 3$(mod 5) and $X \not\equiv 1, 2, 4$(mod 5).*

Proof. Let A be a countably infinite set of atoms with a partition P into sets of size 5 and an enumeration $n \mapsto P_n$. Let $A - 8$ be A with 4 elements removed from P_0 and 4 elements removed from P_1. For even i, let Q_i be the one remaining element from P_i along with 2 elements from P_{i+2} and 2 elements from P_{i+3}. For odd i, Q_i will contain the remaining element from P_i along with 2 from P_{i+1} and 2 from P_{i+2}. Let M be the permutation model generated by P and Q but not their enumerations.

Note that Q breaks up each P block into one distinguished element and 2 pairs.

For even i, we will adopt the notation $T_i = P_i \cup P_{i+1}$.

To show that A is Dedekind finite, assume that $E = \bigcup_{i=0}^{N} T_i$ is fixed. We can still transpose Q_i and Q_{i+1} for any even $i > N$ while fixing E. So A is Dedekind finite.

Clearly P divides A by 5 and Q divides $A - 8$ by 5. Note that $A \equiv 8 \Leftrightarrow A \equiv 3$.

Now assume $A - r$ is divisible by 5 as witnessed by the partition $\mathcal{S}$.

Claim: If two pairs x and y are in the same $\mathcal{S}$ block, then they are in the same T block.

Proof: If false, there exists x^* which is the other pair in x's Q block. There exists an automorphism which transposes X with x^* while Y is fixed. So the $\mathcal{S}$ block contains 6 elements.

Claim: If x is a pair in T_i, and y is a distinguished element in the same $\mathcal{S}$ block, then y is in T_i or T_{i-1}.

Proof: If false, there exists x^* as in the previous claim and so x, x^*, and y are all in the $\mathcal{S}$ block. There also exists y^* in the same T block as y. There exists an automorphism which fixes x and x^* but transposes y with y^*. Therefore the $\mathcal{S}$ block has 6 elements.

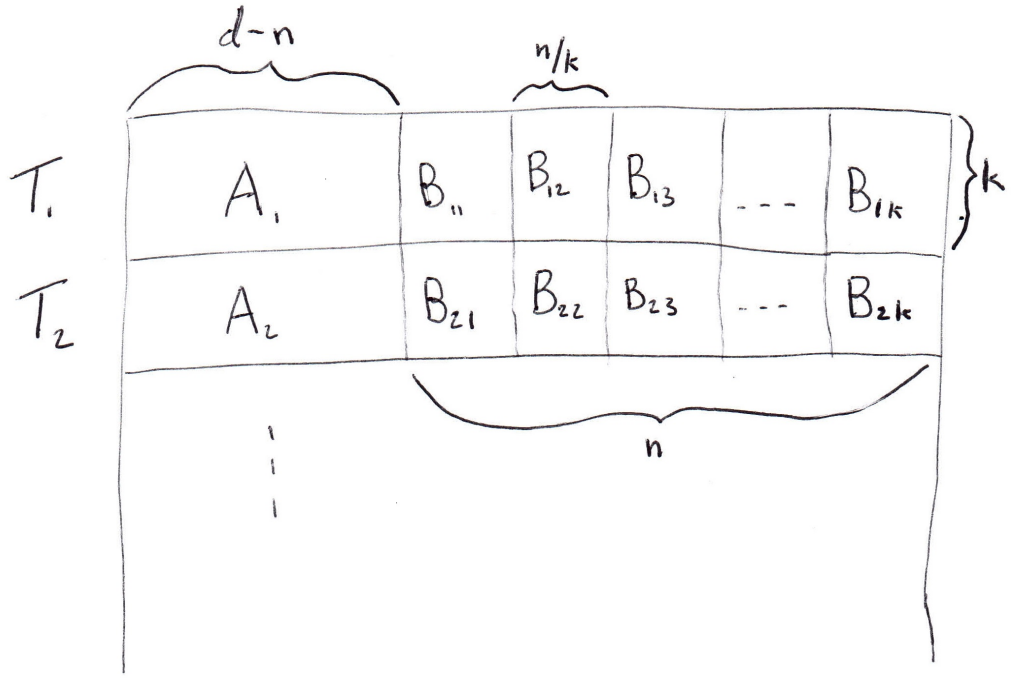
Let $E = \bigcup_{i=0}^{N} T_i$ support $\mathcal{S}$. Fix some even $n >> N$. A pair in T_n will have to be grouped (by $\mathcal{S}$) with another pair from T_n. There will then be another distinguished element from either T_n or T_{n-1}. It is easy enough to see that if the distinguished element is in T_n, then every T_i for $i > n$ will have exactly two $\mathcal{S}$ blocks (case 1). If the distinguished element is in T_{n-1} then every T_i for $i > n$ will contain the pairs from two $\mathcal{S}$ blocks and the distinguished elements from the $\mathcal{S}$ block whose pairs are in T_{i+1} (case 2).

In case 1, there are $10n - r$ elements left after removing the $\mathcal{S}$ blocks above n. Since those $10n - r$ elements must be divisible by 5, $r \equiv 0 \pmod 5$. In case 2, there are $10n - 2 - r$ elements remaining so $r \equiv 3 \pmod 5$. □

From theorem 1.5, we can form a more generalized theorem in a similar manner to theorem 4.1 from [HT].

Theorem 1.6. *If $n < d$, k divides n and $k \geq 2$, then it is consistent to have Dedekind-finite X such that $X \equiv 0$ and $X \equiv kn \pmod d$ while $X \not\equiv r$ for any other $r \neq 0, nk \pmod d$.*

Proof. Let X be a countably infinite set of atoms. P will be a partition of X into rows of d elements. Every k rows will be sectioned into boxes, one A box and k B boxes. The A box will be $(d - n)$ by k and the B boxes will be (n/k) by k as can be seen by the following figure.



T_i will be used to denote the ith row of boxes.

The partition Q of $X - nk$ will contain in each block a B box along with a row of the A box from the preceding T_i. For simplicity say B_{xy} will be grouped (in Q) with the yth row of $A_{(x-1)}$. M will be the permutation model generated by X along with P and Q.

To show that X is Dedekind-finite, take any finite $E \subseteq \bigcup_{i=1}^{N} P_i$. For $n >> N$, P_n meets the same T_i as P_{n-1} or P_{n+1}. Assume without loss of generality that it is P_{n+1}. There exists an automorphism which transposes P_n with P_{n+1} and the two Q blocks that touch P_n and P_{n+1} in the A block while fixing everything else. Thus X is Dedekind-finite.

Now we must show that $X \equiv 0, kn$ and $X \not\equiv r$ for $r \neq 0, kn$. The partition P shows that $X \equiv 0$, while Q shows that $X \equiv nk$. In order to see that X has no other congruences, assume $\mathcal{S}$ is a partition of $X - r$ into sets of size d, and $E = \bigcup_{i=0}^{N} T_i$ is a support of E.

Case 1: $n > d/2$

Clearly (outside of E) any two elements of the same row in the same box must share an $\mathcal{S}$ block. In light of this fact, I will be referring mostly to A-rows and B-rows throughout the proof.

Claim: Outside of E, if two B-rows are in the same $\mathcal{S}$ block, they are in the same T block.

Proof: Assume this is false. Each of the two B-rows can be permuted with $k-1$ B-rows while the other is fixed. Thus there are $2k$ B-rows in the same $\mathcal{S}$ block and so there are $2k * (n/k) = 2n > d$ elements in a block.

Claim: Outside of E, if an A-row is in T_i, and it is in the same $\mathcal{S}$ block as a B-row, then the B-row is in T_i or T_{i+1}.

Proof: Assume the claim is false, then the A-row can be transposed with $k-1$ other A-rows while fixing the B-row. Similarly, the B-row can be transposed with $k-1$ B-rows while fixing the A-row. So there would be $k(d-n) + k(n/k) = kd - (k-1)n > d$ elements in one $\mathcal{S}$ block.

Claim: Outside of E, if an A-row is in T_i and it is in the same $\mathcal{S}$ block as a B-row in T_k, then any other A-row in T_i must only be grouped (in $\mathcal{S}$) with other B-rows from T_k.

Proof: Assume the claim is false. There exists an automorphism which transposes the two A-rows while fixing one of the B-rows. Thus all of the rows mentioned are in the same $\mathcal{S}$ block contradicting the first claim.

This leaves two subcases.

Subcase 1: Each $\mathcal{S}$ block consists of an A-row and k B-rows all from the same T_i

In this case there are $dm - r$ elements in $X - r$ which are also in the support. These elements are divisible by d as witnessed by $\mathcal{S}$. So $r \equiv 0 \pmod{d}$.

Subcase 2: Each $\mathcal{S}$ block consists of an A-row from some T_i and k B-rows from T_{i+1}

In this case consider all of the $\mathcal{S}$ blocks touching the support. There are $dm + nk - r$ of these and they are divisible by d. Thus $r \equiv nk \pmod{d}$

Case 2: $n = d/2$

This case is proved in [HT]

Case 3: $n < d/2$

As in case 1 any two elements (outside of E) from the same row and box must be in the same $\mathcal{S}$ block. Also, there clearly can not be two A-rows in the same $\mathcal{S}$ block because together they have more than d elements. Thus each A-row is grouped with

Claim: Outside of E, if an A-row is in T_i, and it is in the same $\mathcal{S}$ block as a B-row, then the B-row is in T_i or T_{i+1}.

Proof: Assume the claim is false, then the A-row can be transposed with $k-1$ other A-rows while fixing the B-row. Similarly , the B-row can be transposed with $k-1$ B-rows while the A-row is fixed. So there would be $k(d-n)+k(n/k) = kd-(k-1)n > d$ elements in one $\mathcal{S}$ block.

Claim: Outside of E, if a B-row is in T_i and it is in the same $\mathcal{S}$ block as an A-row. Any other B-rows in that $\mathcal{S}$ block must be in T_i.

Proof: Assume the claim is false. Then each B-row can be transposed with $k-1$ other B-rows while fixing the A-row. Therefore there would be $(d-n)+2k(n/k) > d$ elements in an $\mathcal{S}$ block.

Claim: Outside of E, if an A-row is in T_i and it is in the same $\mathcal{S}$ block as a B-row in T_k, then any other A-row in T_i must only be grouped (in $\mathcal{S}$) with other B-rows from T_k.

Proof: Assume the claim is false. There exists an automorphism which transposes the two A-rows while fixing one of the B-rows. Therefore both A-rows are in the same block which is a contradiction.

It is not hard to see that this gives the same two subcases as case 1 and therefore comes to the same conclusion that $r = 0 \pmod{d}$ or $r = nk \pmod{d}$.

□

I do not feel that it is important to reproduce it here but it is easy to see how the method of case 3 in the preceding theorem can be applied to prove that theorem 4.1 from [HT] works for $n > d/2$.

Theorem 1.7. *For any natural number d, natural numbers $n_0, n_1, ..., n_l$, and natural numbers $k_0, k_1, ..., k_l$ such that each triple (d, n_i, k_i) satisfies the hypothesis of theorem 4.1 from [HT] or theorem 1.6, then it is consistent to have Dedekind-finite X such that X is congruent (mod d) to 0, $k_i n_i$ for all i and any number which is a sum of the different $k_i n_i$. X is not congruent to anything else.*

Proof. For each i, X_i will be a countable set of atoms with the structure defined in the appropriate preceding theorem. Let $X = \bigcup_{i \leq l} X_i$. Let M be the permutation model generated when considering the permutations which preserve the structure on the X_i. Clearly X is congruent (in M) to the numbers that we want. All that is left is to prove that it is not congruent to anything else. Assume that Q is a partition dividing $X - r$ into sets of size d and E supports Q.

Claim: Outside of E, an A-row or B-row can not be in the same $\mathcal{S}$ block as an A-row or B-row from a different X_i.

Proof: Assume they share an $\mathcal{S}$ block. The second one may be transposed with infinitely many rows in its X_i while the first is fixed thus there would be far too many elements in the same $\mathcal{S}$ block.

In light of the claim, the possible structure of the X_is, outside of E, is the same as in the preceding theorems. Thus the $\mathcal{S}$ blocks touching E contain $10m + k - r$ elements where k is a sum of the $k_i n_i$. □

This gives a somewhat general picture of non consecutive congruences but it is not complete. I believe the case of $X \equiv 0, 3 \pmod 8$ is still open.

2 Partition Structures

Theorem 2.1. *(Modified from 6.17 of [BBH] using the same notation) Let $d = p_1 + ... + p_k$ with all $p_i \geq 2$, and let $G_1, ..., G_k$ where $G_i \subseteq Sym(p_i)$ have the property, $(\forall a \in p_i), (\exists g \in G_i) g(a) \neq a$, let $\mathcal{S}$ be the d-species where a structure on F is a partition of F into pieces $F_1, ..., F_k$ of sizes $p_1, ..., p_k$ together with a $\mathcal{B}_{G_i}$ structure on each piece corresponding to p_i. If the greatest common divisor of $p_1, ..., p_k$ is 1, then it is consistent to have a Dedekind-finite X such that $X \pm n$ are divisible by d with structure $\mathcal{S}$ for all natural numbers n.*

Proof. Since the greatest common divisor is 1, we know that $1 = p_1 r_1 + ... + p_k r_k$ where the r_i's are integers. We can rearrange and renumber this equation to get

$$1 + p_1(-r_1) + ... + p_l(-r_l) = p_{l+1} r_{l+1} + ... + p_k r_k$$

where all the terms are positive.

On a countably infinite set A of atoms, put the following structure:

1. a partition into pieces $P_n (N \in \mathbb{N})$ of size d,

2. for each piece P_n, a partition into subpieces $P_{n,i}$ of sizes $p_i (1 \leq i \leq k)$,

3. the function $(n, i) \mapsto P_{n,i}$ labeling the subpieces, and

4. for each Pn, i, a function $b_{n,i} : p_i \to P_{n,i}$

Since $b_{n,i}$ is a function from p_i to $P_{n,i}$, $[b_{n,i}]_{G_i}$ is a $\mathcal{B}_{G_i}$ structure on $P_{n,i}$. Let $\mathfrak{A}$ be the set A with 1, 2, 3, and the $[b_{n,i}]$'s generated from 4. Thus an automorphism of $\mathfrak{A}$ must map each subpiece to itself, preserving $[b_{n,i}]$ Let M be the resulting permutation model. Since the partition into pieces P_n and the assignment, to each P_n, of its partition into subpieces and the $\mathcal{B}_{G_i}$ structures of the subpieces are symmetric, A is divisible by d with structure $\mathcal{S}$ in M.

We next show that $A + 1$ has the same divisibility property. Let X be the element added to A to form $A+1$. Let Z be the set consisting of x and, for each $i \leq l$, the elements of $P_{n,i}$ for the first $(-r_i)$ values of n. The cardinality of this set is, by our choice of the r_i's, $p_{l+1}r_{l+1} + ... + p_k r_k$, so we can partition it into r_j blocks of size p_j for $j = l + 1, ..., k$, and we can fix (in M) a $\mathcal{B}_{G_i}$ structure on each p_i size block (finitely many blocks so AC need not be not used). It is easy to rearrange these new blocks along with the surviving old subpieces into a partition $\mathcal{Q}$ of $A + 1$ into pieces of size d equipped with $\mathcal{S}$-structures. The nth block of $\mathcal{Q}$ consists of, for $i \leq l$, the blocks $P_{(n+r_i),i}$ and, for $l + 1 \leq k$, the blocks $P_{(n-r_i),i}$, where for positive r_i the subpieces with negative first subscript mean the blocks in Z newly created by the rearrangement.

To show the same divisibility property for $A + n$ with $n > 1$, use the same technique, but rearrange n times as many subpieces. For $A - n$, add a multiple of d to return to the case where points are added to A. Thus, all $A \pm n$ are divisible by d with $\mathcal{S}$-structure.

It remains to show that A is Dedekind-finite in M. If E is a finite subset of A, E intersects with only finitely many P_n. Call the union of those P_n the P closure of E or $P(E)$. We will show that for any $t \in A$ which is not in $P(E)$ there exists an automorphism π which fixes all of $P(E)$ but moves t. Let $t = b_{n,i}(c)$. By the hypothesis, we can choose $g \in G_i$ such that $g(c) \neq c$. Choose a permutation π of A which sends $b_{n,i}(c) \mapsto b_{n,i}(g(c))$, and fixes averything else in A (outside of $P_{n,i}$). This preserves the $[b]_{G_i}$ structure on $P_{n,i}$ because it sends it to $[\pi \circ b]_{G_i} = [b \circ g]_{G_i}$ which equals $[b]_{G_i}$ because $g \in G_i$. π clearly preserves the partition structure. Thus A is Dedekind-finite. □

Example:

Probably the most simple example is the case where there is no structure on each piece. In other words each piece may be permuted in any way. While the theorem applies to this example it is not very interesting because it was

already covered by theorem 6.11 of [BBH]. A more interesting example is where each p_i is odd and ≥ 5. Then the structure of each subpiece is a partition into two cyclic orderings. By the theorem we have divisibility while this structure does not fall under theorem 6.11 or 6.17 of [BBH].

3 References

[BBH] D. Blair, A. Blass, P. Howard, Divisibility of Dedekind Finite Sets, Journal of Mathematical Logic 5 (01):49-85 (2005)

[HT] J. Holler, M. Tanzer, Non-consecutive Congruences of Dedekind-finite Sets